

V6 vLLM Characterization Dashboard

Deep Audit & UI Remediation Specification

Kimi-Linear-48B-A3B-Instruct on NVIDIA RTX PRO 6000 Blackwell Server Edition

Purpose Turn the current dashboard into an architect-grade, provenance-backed inference characterization UI. The report specifies exactly what must be retained, corrected, added, removed, or marked unresolved. Special focus: 1M context behavior, scale-out TP/PP topology, serving capacity, scheduler/KV dynamics, and TTFT/TPOT decomposition.

Evidence discipline Qualification values were previously raw-validated from uploaded V6 qualification logs. The latest full-production dashboard contains additional V6 values; where the underlying full production raw folders are not locally attached, this report treats those values as "current UI values" pending raw re-validation. No synthetic or interpolated value should be labeled measured.

Prepared for: Architecture / vLLM / GPU systems review

Date: 21 September 2026

1. Executive assessment

Overall assessment The dashboard architecture is strong: Executive, Scale-Up, Scale-Out, Long Context, Scheduler & KV, Profiler, and Evidence are the correct seven views. However, it must not be shipped unchanged. Several charts contain hard-coded/interpolated values presented as measured, hardware labels still contain legacy Ada/48GB/PCIe Gen4 assumptions, scheduler/KV cards conflict with the Evidence table, and the Profiler tab currently presents illustrative decomposition percentages as measured runtime attribution.

Priority	Area	Required outcome
P0	Data integrity	Remove hard-coded "measured" arrays; every point must resolve to a run/evidence row.
P0	Hardware identity	Replace RTX 6000 Ada / 48GB / PCIe Gen4 with RTX PRO 6000 Blackwell Server Edition / 96GB GDDR7 / PCIe Gen5.
P0	1M context	Keep measured 1M results, but separate baseline feasibility from full 1M matrix coverage. Offload and multi-node 1M remain absent/not surfaced.
P0	Scale-out	Correct TP16 and forced TP4/PP2 topology diagrams; remove universal winner/causal claims unless backed by matched raw traces.
P0	Scheduler/KV	Replace synthetic KV utilization and low-N percentile claims with actual evidence rows and sample-count gates.
P0	Profiler	Keep the equation; remove fabricated 48/22/8/14/8/-12.6% shares until timeline attribution exists.
P0	Coverage/provenance	Show configured/completed/failed/gated/not-run counts and suite revision; current dashboard has 59 evidence rows vs 64 configured rows in archived V6 plus one later extra 1M c4 row.

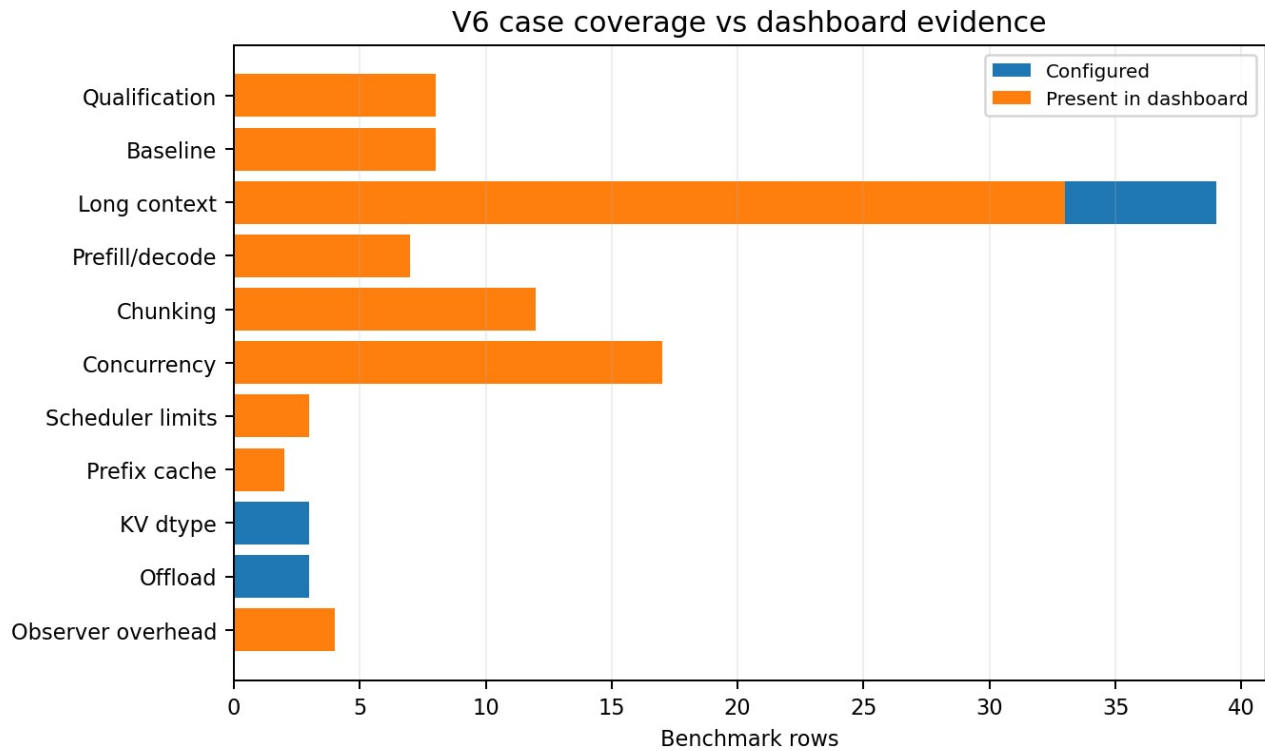


Figure 1. Coverage in the archived V6 case matrix versus rows present in the current dashboard.

Interpretation: The dashboard is complete for qualification/baseline/chunking/concurrency/scheduler/prefix/observer groups, but the FP8-KV and native-offload groups are absent. The long-context aggregate therefore remains incomplete.

2. Scope, source hierarchy, and validation status

- Primary UI source reviewed: MASTER_CHARACTERIZATION_DASHBOARD.html (2,523 lines; seven tabs).
- V6 archived case definition reviewed: v5_vllm_profiling_suite_v6/rtx_g4_smoke_v5/10_vllm_surrogate_cases.json and 10b_vllm_multi_node_cases.json.
- Raw evidence locally available and previously audited: V6 qualification logs, manifests, benchmark JSON, server logs, Prometheus snapshots, and GPU telemetry.
- Historical V5 Option A/B/C archives were previously reviewed to identify profiler and multi-node pitfalls.
- External hardware check: NVIDIA specifies RTX PRO 6000 Blackwell Server Edition as 96GB GDDR7, PCIe Gen5 x16, 1597 GB/s memory bandwidth.
- vLLM semantics check: request-rate and max-concurrency are separate; max_num_seqs is an independent scheduler limit; explicit KV cache memory overrides gpu_memory_utilization; KV offload size is total across TP ranks in vLLM 0.29.0.

Raw-validation boundary The current HTML contains full-production values beyond the uploaded qualification package. Those values are useful for UI auditing, but must be linked to the final raw run manifest before the UI can label them as measured. Scale-out and profiler numbers require especially strict provenance.

3. What the current UI gets right

- The seven-tab information architecture is appropriate and should be retained.
- The dashboard separates scale-up, scale-out, long-context, scheduler/KV and profiler concerns instead of collapsing them into one benchmark page.

- The Evidence tab is the right concept for auditability; it should become the actual source-of-truth layer for every chart.
- The Executive page correctly tries to expose TTFT, TPOT, throughput, topology and runtime reserve rather than only GPU utilization.
- Qualification results show the harness can capture actual input-token lengths, TP4/TP8 metrics, KV peak, preemption state and warmup-separated benchmark results.
- The runtime accounting equation is directionally correct as a framework if overlap and residual are treated explicitly rather than force-fitting percentages.

4. Global UI corrections - implement before sign-off

Pri	Issue	Current risk	Exact change
P0	Hardware identity	Legacy Ada/48GB/Gen4 text exists.	Use detected GPU model + official Blackwell metadata: 96GB GDDR7, PCIe Gen5. Never hard-code capacity.
P0	Dataset time/status	Header shows Apr 27, 2025 and "Live".	Show latest run timestamp and dashboard build time. Only show Live when polling active runs.
P0	Evidence classes	Model metrics and hardware metrics are mixed.	MEASURED-48B=model/vLLM; MEASURED-GCP-HW=hardware/fabric; DERIVED=calculated; LOCAL-REAL=local K3 only; UNRESOLVED=missing.
P0	Data binding	Chart arrays are embedded in JS.	Every point must be generated from result JSON with run_id/source_file/sample_count.
P0	Missing values	Some charts interpolate or fill absent cells.	Render NOT RUN / NOT CAPTURED; never 0 or plausible filler.
P0	Annotation language	Some cards say best/recommended/proves.	Use Observation -> Interpretation(confidence) -> Decision(workload-scoped) -> Next evidence.
P0	Revision drift	Evidence includes 1M c4 not present in archived V6 case file.	Expose suite_version/git_SHA/case_manifest hash in header and Evidence tab.

5. Executive tab - precise changes

Action	Component	Instruction
KEEP	TP4 decode qualification card	Useful if explicitly scoped to matched measured points.
CHANGE	TP8 long-prefill card	Replace qualification-only wording now that 1M baseline is present; state TP8 lower TTFT at 512K and 1M c1 in current dashboard, but do not imply universal winner.
CHANGE	8/8 qualification status	Replace top-level status with V6 coverage summary. Keep 8/8 as sub-card.
CHANGE	Serving envelope heatmap	Use only measured cells. Remove synthetic 32K/64K/256K values and unavailable concurrency cells.
CHANGE	Runtime reserve bar	Replace quantitative percentages with component ledger until profiler attribution is raw-validated.

ADD	Coverage strip	Configured / completed / failed / safety-skipped / not-run by test group.
ADD	Data freshness	Last run, suite revision, model revision, vLLM version, profiler status, scale-out raw-validation status.

6. Scale-Up tab - measured behavior vs UI overreach

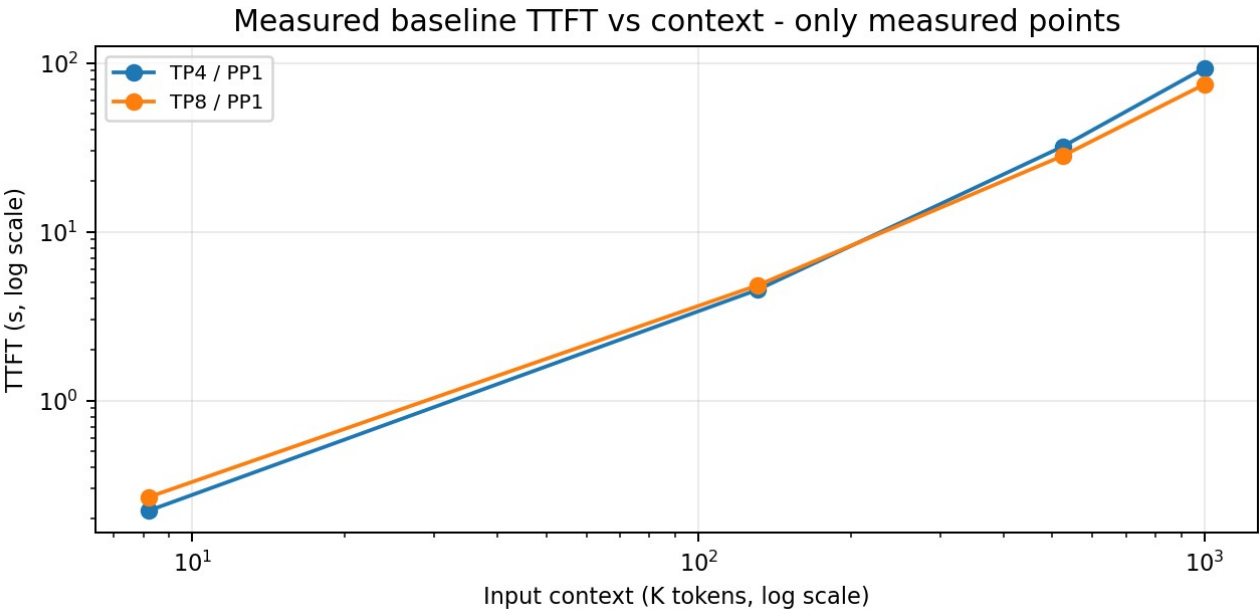


Figure 2. Baseline TTFT uses only measured 8K, 128K, 512K and 1M points.

Interpretation: The current UI should not insert 32K, 64K or 256K points into a chart labeled measured. If interpolation is useful, render it as a dashed DERIVED line.

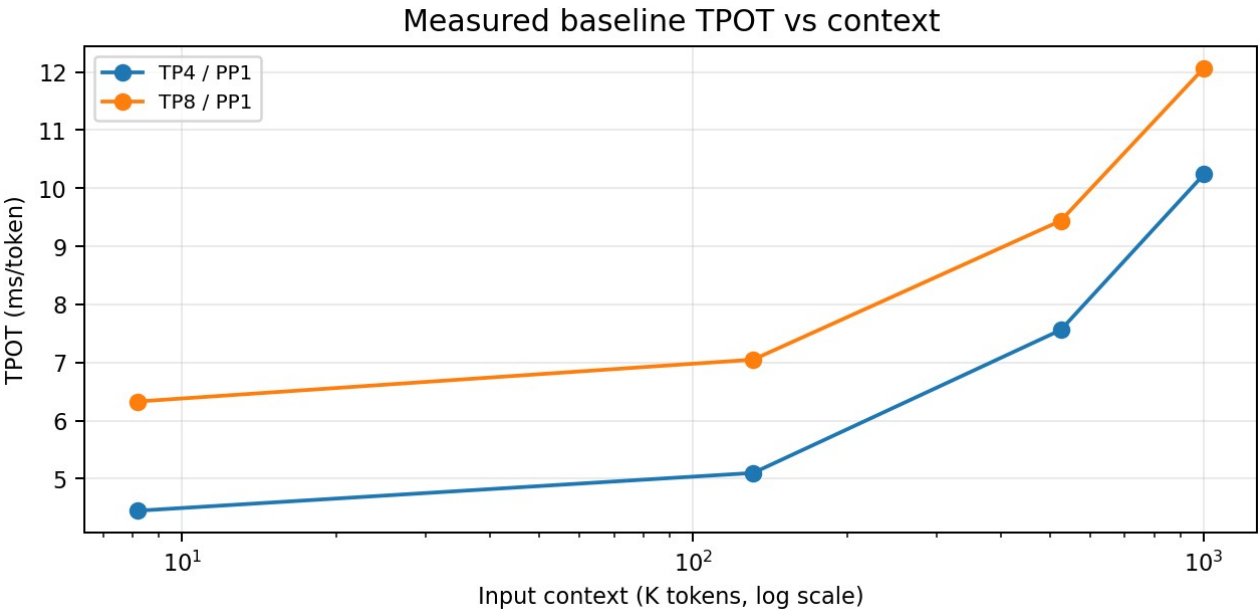


Figure 3. Baseline TPOT by context.

Interpretation: TP4 retains lower TPOT than TP8 at the measured baseline points, while TP8 has lower TTFT at 512K/1M. This is a prefill-vs-decode topology trade-off, not a universal TP winner.

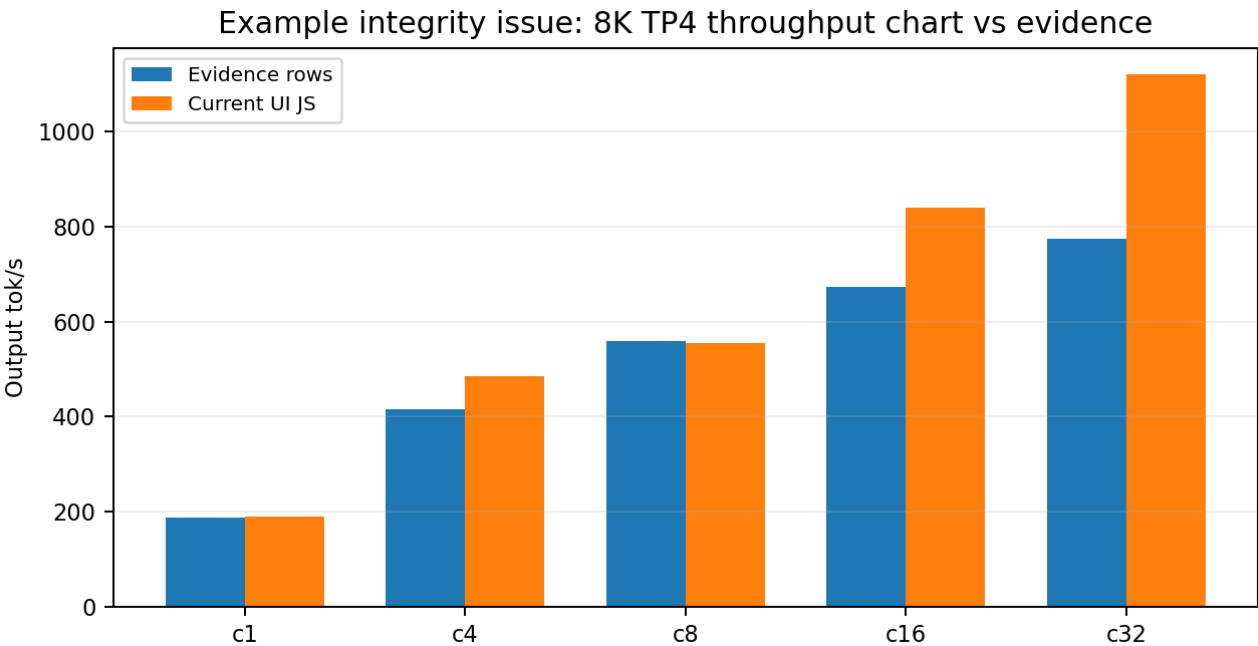


Figure 4. Example of current UI hard-coded 8K TP4 throughput diverging from the Evidence table.

Interpretation: At high concurrency, the dashboard JS overstates throughput relative to its own Evidence rows. All graphs must bind to the evidence database, not embedded arrays.

Pri	Panel	Required change
P0	TTFT curve	Remove invented 32K/64K/256K measured markers. Measured = 8K/128K/512K/1M.
P0	"~350K crossover"	Change to "observed crossover bracket: between 128K and 512K". Fitted crossover must be DERIVED.
P0	TPOT vs concurrency	Do not synthesize TP8 c2/c4/c16/c32. Separate TP4 full sweep from matched TP4-vs-TP8 points.
P0	Throughput vs concurrency	Use exact Evidence rows. Add N, output length and closed-loop label to tooltip.
P0	AllReduce graph	Source from V4/NCCL or profiler. Badge as MEASURED-GCP-HW or profile-derived; distinguish time/algbw/busbw.
P1	Pareto view	Add throughput vs TPOT scatter by context/concurrency. This exposes the latency-throughput frontier.
P1	Capacity knee	Mark point where throughput gain flattens while TPOT/TTFT rises sharply; do not call it max users.

7. Serving-capacity analysis the UI should expose

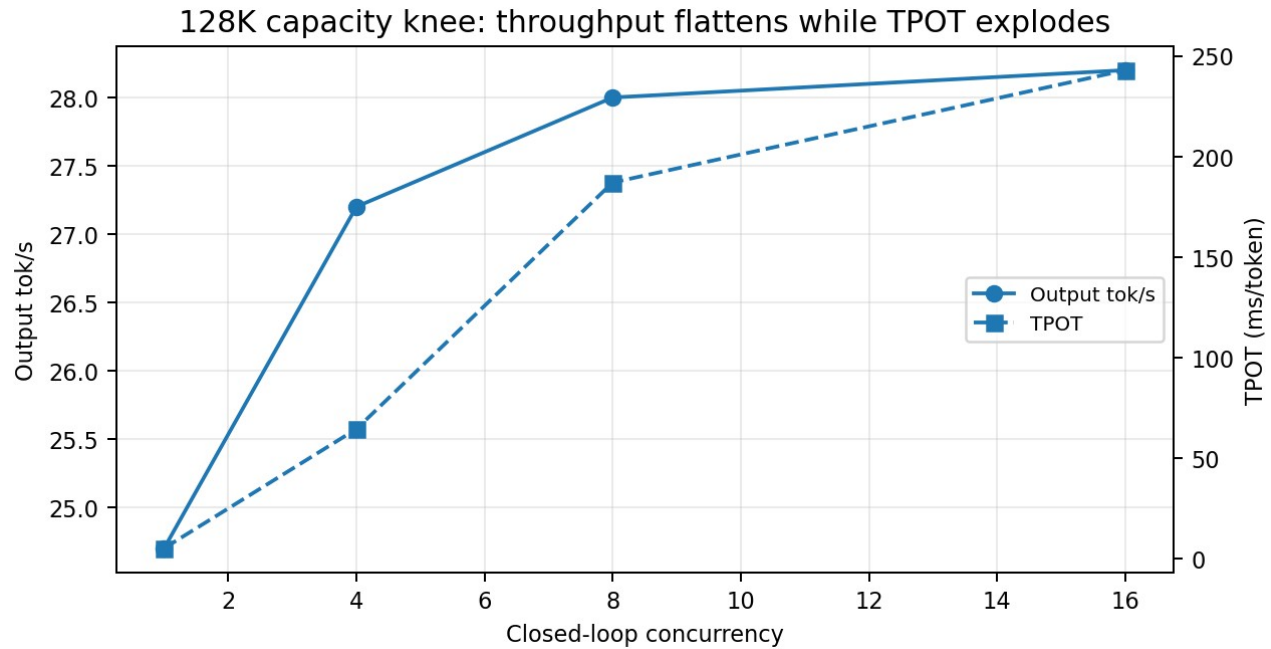


Figure 5. 128K closed-loop capacity knee from dashboard Evidence rows.

Interpretation: Throughput increases only from 24.7 to 28.2 tok/s between c1 and c16, while TPOT rises from 5.11 to 242.85 ms/token. The UI should highlight this as a latency-throughput trade-off, not successful scaling.

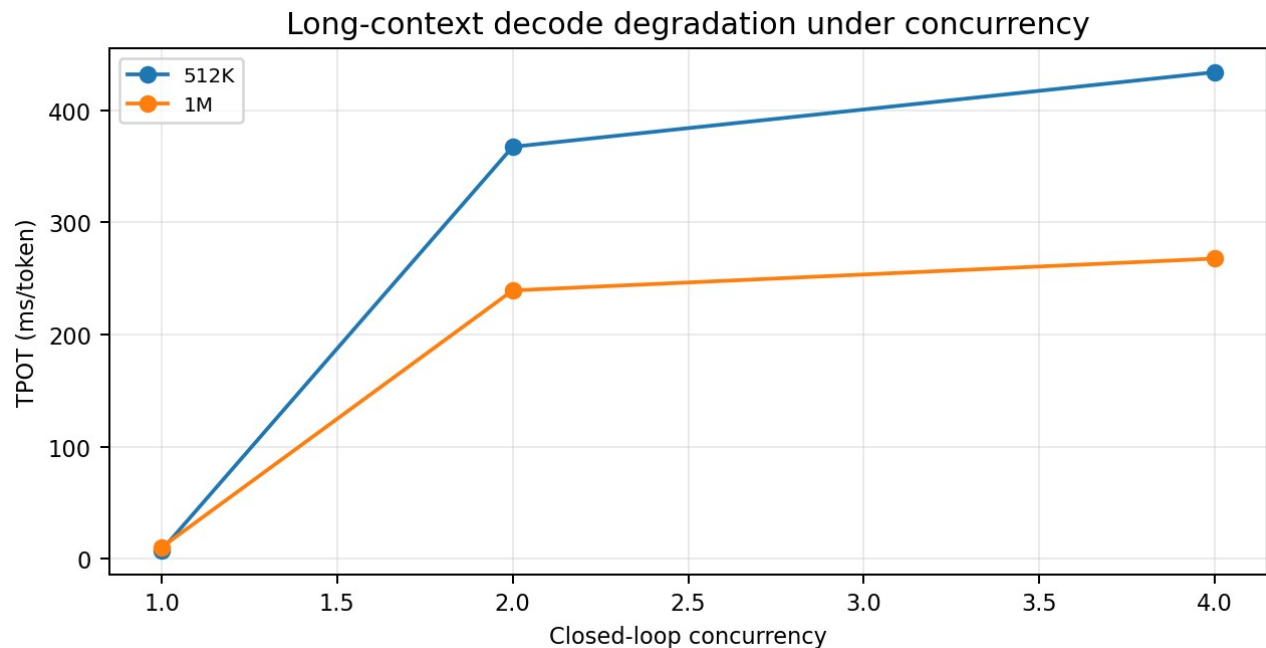


Figure 6. Long-context TPOT degradation under concurrency.

Interpretation: At 512K and 1M, c2/c4 do not increase the displayed output throughput materially, but TPOT increases by hundreds of milliseconds. This is a key serving-capacity result and should be above the fold in Scheduler & KV / Long Context.

Critical interpretation Closed-loop concurrency is not "number of users supported." vLLM bench serve distinguishes max-concurrency from request-rate; request-rate=inf sends all requests immediately. Capacity must be

expressed as a workload-specific envelope across context, output length, concurrency, offered RPS, queue, TTFT/TPOT SLO and KV/preemption state.

8. 1M context - deep audit and required UI treatment

1M status The dashboard contains substantial single-node 1M evidence. It does not establish a complete 1M matrix. Baseline TP4 c1 and TP8 c1 are present; TP4 prefill-focused, chunk-size, and closed-loop c1/c2/c4 are present. The archived V6 offload 1M case is absent from the dashboard; multi-node 1M is not part of the multi-node case matrix.

Case	Bench	TP/PP	Conc.	TTFT	TPOT ms	Out tok/s	KV peak
tp4_context_base line	1m_c1	TP4 / PP1	c1	93.38s	10.24	0.30	12.3%
tp8_context_base line	1m_c1	TP8 / PP1	c1	74.85s	12.07	0.40	12.2%
tp4_prefill_focus	1m_prefill	TP4 / PP1	c1	93.35s	10.64	0.00	12.3%
tp4_chunk4k	1m_c1	TP4 / PP1	c1	122.10s	10.28	0.10	12.2%
tp4_chunk8k	1m_c1	TP4 / PP1	c1	93.38s	10.37	0.20	12.3%
tp4_chunk16k	1m_c1	TP4 / PP1	c1	89.16s	10.26	0.20	12.5%
tp4_closedloop_1 m	c1	TP4 / PP1	c1	93.46s	10.20	0.30	12.3%
tp4_closedloop_1 m	c2	TP4 / PP1	c2	150.65s	239.25	0.30	15.5%
tp4_closedloop_1 m	c4	TP4 / PP1	c4	231.50s	267.69	0.30	15.5%

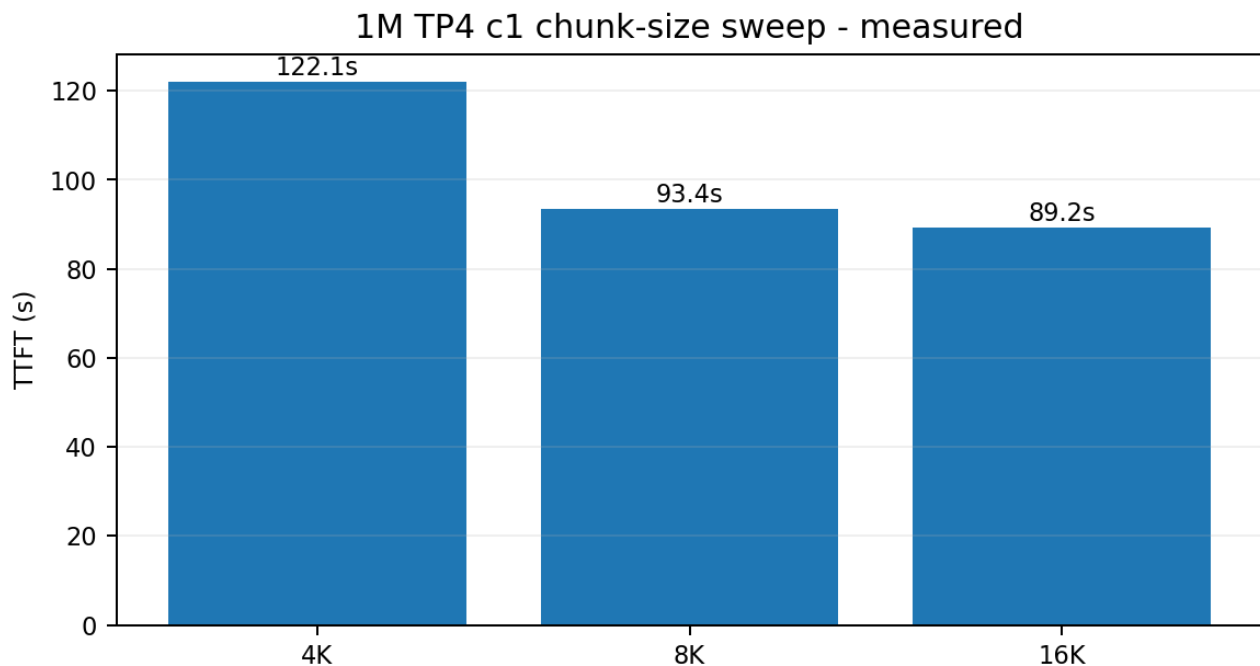


Figure 7. 1M TP4 c1 chunk-size sweep.

Interpretation: In the current evidence rows, 16K has the lowest TTFT (89.16s), 8K is 93.38s, and 4K is 122.10s. Therefore the current "8,192 sweet spot" claim is too strong without concurrent-decode/SLO context.

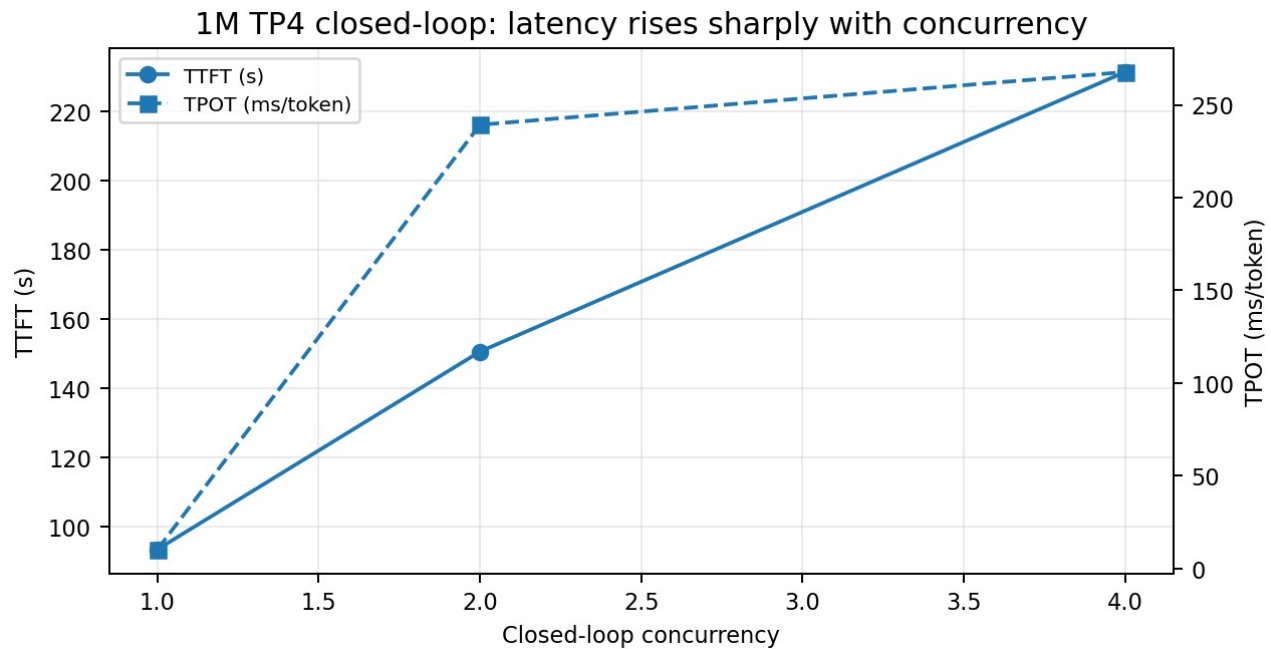


Figure 8. 1M TP4 closed-loop concurrency.

Interpretation: TTFT rises from 93.46s at c1 to 150.65s at c2 and 231.50s at c4; TPOT rises from 10.20ms to 239.25ms and 267.69ms. This must be one of the headline long-context charts.

Pri	1M item	Instruction
P0	Feasibility KPI	Replace "100% Passed" with exact coverage: TP4/TP8 c1 baseline + TP4 c1/c2/c4 closed-loop completed.
P0	Chunk KPI	Replace "8K sweet spot" with measured finding: 16K lowest 1M c1 TTFT among 4K/8K/16K. Add workload/SLO caveat.
P0	Prefix 1M	Do not label 14.4s at 90% hit as measured unless a real 1M prefix run exists. Use 128K/512K cold-vs-repeat; 1M projection = DERIVED.
P0	KV memory	Do not convert 12.3% utilization into 12.3GB. Show % unless configured KV bytes are verified.
P0	Context curve	Remove synthetic 32K/64K/256K measured markers.
P1	Regime-change panel	Plot per-prefill-chunk iteration time vs accumulated context together with GPU util, KV%, scheduler time, offload bytes/time and TP/NCCL contribution.
P1	1M boundary	Keep 1,000,000-token nominal test distinct from 1,048,576 model maximum.
P1	Missing coverage	Show 1M offload as NOT RUN/NOT SURFACED; show 1M multi-node as NOT CONFIGURED unless added later.

9. Scale-Out tab - deep correction specification

Important The Scale-Out tab was explicitly reviewed. Its topology intent is good, but several labels and causal claims are too strong. The current UI values below should remain "current UI values pending raw production validation" until each row is linked to its run manifest, rank placement, network fingerprint and source JSON.

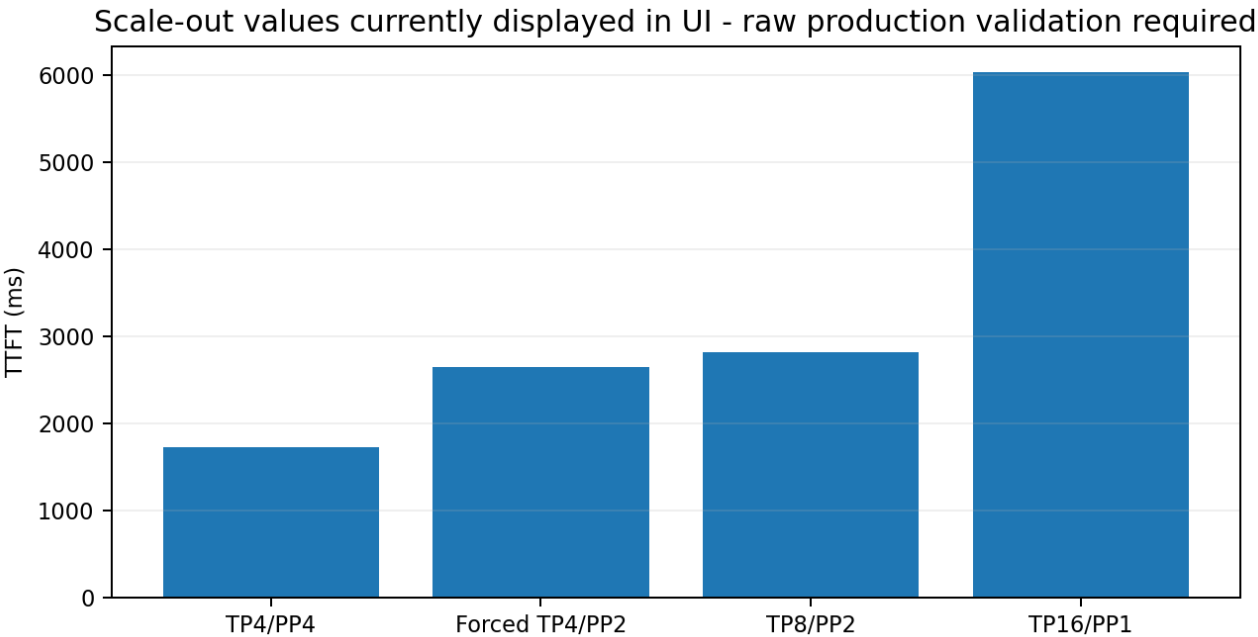


Figure 9. 128K scale-out TTFT values currently displayed by the UI.

Interpretation: These values are useful for comparison only after raw multi-node validation. The label must be MEASURED-48B when validated, not MEASURED-GCP-HW.

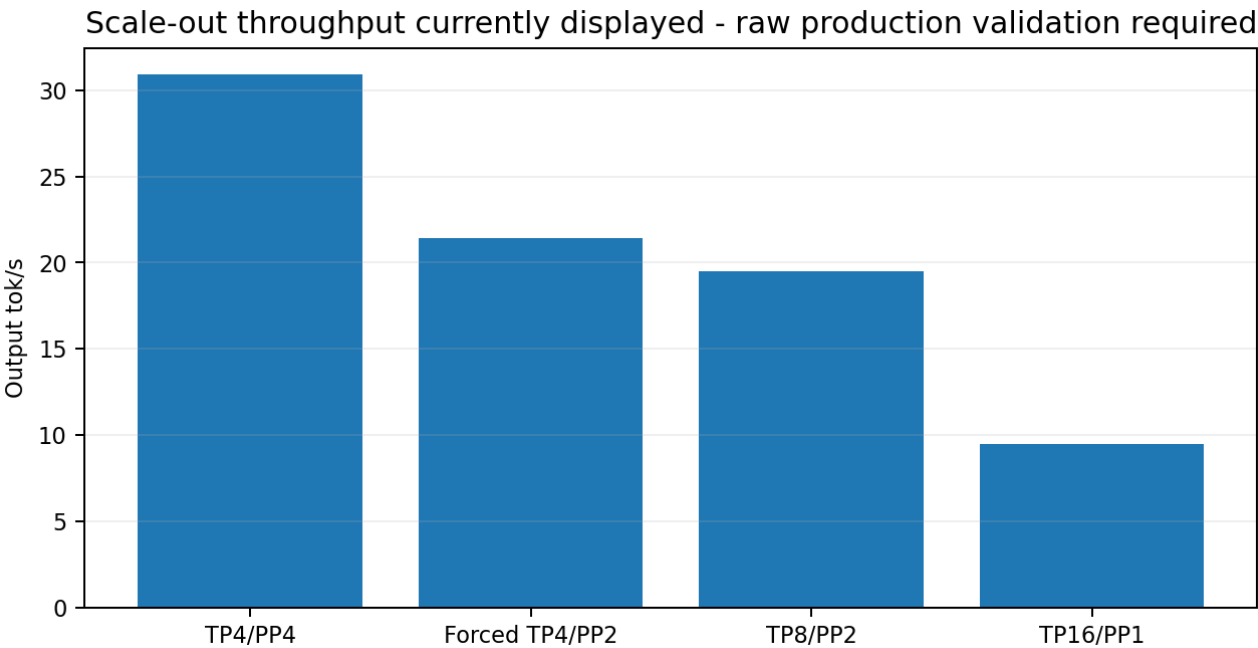


Figure 10. Scale-out throughput values currently displayed by the UI.

Interpretation: Do not infer that PP "masks" network latency from throughput alone. Confirm with NCCL and PP-stage timelines.

Pri	Scale-out item	Instruction
P0	TP16/PP1 diagram	Must show a TP16 group spanning Node0 and Node1; it is not "within nodes".
P0	Forced TP4/PP2 diagram	Must show one TP4 stage on Node0 -> remote PP boundary -> one TP4 stage on Node1. Do not draw a split cross-node TP ring.

P0	Top Distributed Topology	Replace universal winner with "lowest TTFT / highest throughput in matched 128K c1 test".
P0	Causal language	Replace "Pipeline Parallelism Superiority" and "proves pipeline stages mask latency" with observation + medium-confidence interpretation + trace requirement.
P0	Network traffic	Do not state 22.4GB/s saturation/TCP queues unless bytes/s, link capacity, retries/queue data prove it.
P0	Evidence badge	TTFT/TPOT/tok/s = MEASURED-48B; network/MTU/iperf/NCCL primitive = MEASURED-GCP-HW.
P1	Context selector	128K and 512K for TP4/PP2, TP8/PP2, TP4/PP4 where run; TP16/PP1 512K = NOT RUN unless executed.
P1	Placement audit	Render PP rank -> node -> GPU UUID -> PCI bus -> NUMA; PASS/FAIL placement validation.
P1	Per-node balance	Show per-GPU mean/p95 utilization, memory, power and coefficient of variation; a range alone is insufficient.
P1	Network fingerprint	Expose node zones, interface, MTU, RTT, iperf, cap, NCCL transport/plugin state beside each topology.
P1	Topology decision view	Map workload regime to measured tradeoffs. No universal "recommended" label without an explicit objective/SLO.

9.1 Correct topology semantics to show in UI

TP16 / PP1: Node0 GPU0-7 <==== cross-node TP collectives =====> Node1 GPU0-7

TP8 / PP2: Node0 [TP8 stage0] ---- PP boundary ----> Node1 [TP8 stage1]

TP4 / PP4: Node0 [TP4 stage0][TP4 stage1] ---- PP ----> Node1 [TP4 stage2][TP4 stage3]

Forced TP4 / PP2: Node0 visible GPU0-3 [TP4 stage0] ---- PP ----> Node1 visible GPU0-3 [TP4 stage1]

10. Scheduler & KV tab - corrections and stronger analysis

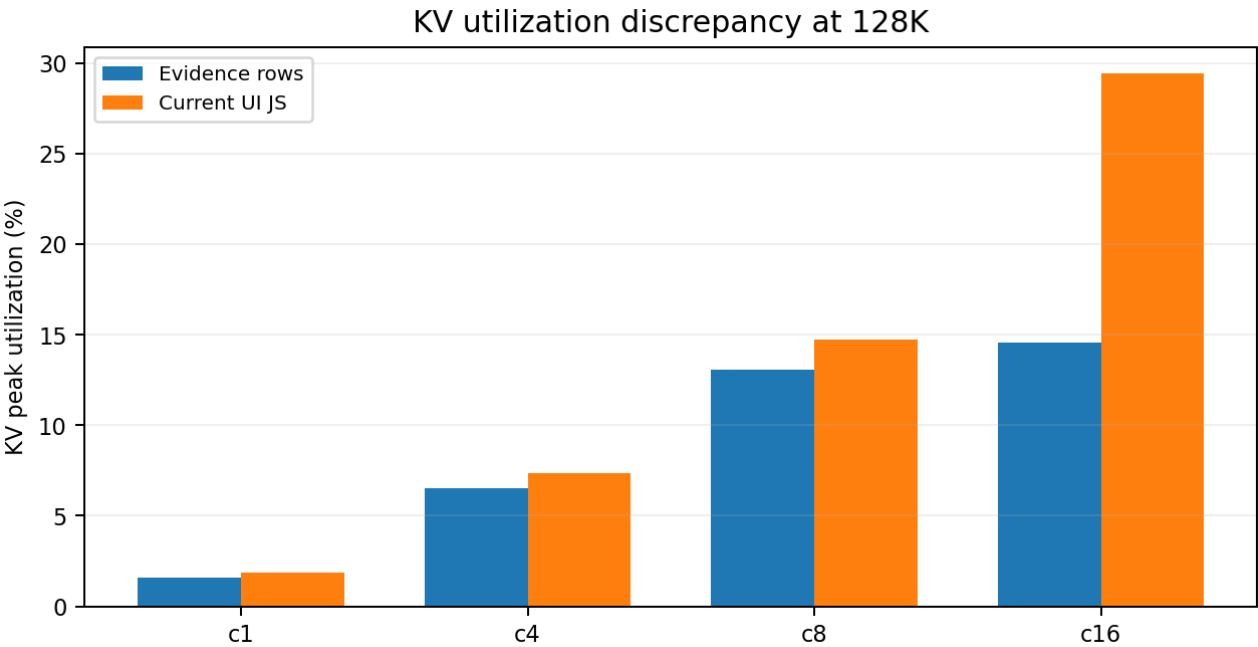


Figure 11. Current KV-utilization chart conflicts with the Evidence table.

Interpretation: At 128K c16 the current evidence row reports 14.6% KV peak, while the current chart logic scales to 29.44%. The chart must use exact kv_peak_pct rows, not synthetic multiplication.

max_num_seqs	TTFT	TPOT ms	Out tok/s	KV peak
4	88.007s	433.81	2.0	12.9%
8	87.994s	433.76	2.0	12.9%
16	87.975s	433.77	2.0	12.9%

Measured scheduler-limit result At 512K c4, max_num_seqs 4/8/16 produces nearly identical TTFT (~87.97-88.01s), TPOT (~433.76-433.81ms) and KV peak (12.9%) in the dashboard evidence. This is a valuable result: within this tested range, max_num_seqs was not the dominant limiter for that workload.

Pri	Scheduler/KV item	Instruction
P0	KV utilization	Use evidence row values only; 1M c4 is 15.5% in the current Evidence table, not 49.2%.
P0	Scheduler overhead	Show <0.04ms only if parsed EngineCore iteration CPU time directly supports it; otherwise NOT CAPTURED.
P0	Fragmentation	Show fragmentation only with a real fragmentation metric; block size alone does not prove <1.8%.
P0	Percentiles	P95 requires >=20 samples; P99 >=100. Low-N long-context runs should show individual samples/min/median/max.
P1	max_num_seqs chart	Promote the 512K c4 4/8/16 sweep as a dedicated chart and learning.
P1	Capacity heatmap	Axes: context x closed-loop concurrency; cell color = TPOT or TTFT; overlay output tok/s and KV%; gray = not run.
P1	Open-loop	Add request-rate vs achieved

throughput/queue/TTFT page once RPS runs exist. Keep external users distinct from max concurrency.

10.1 Prefix-cache and observability panels

Case	Input	Aggregate TTFT	TPOT ms	Out tok/s	KV peak
tp4_prefix128k	131,328	0.911s	5.31	80.7	1.9%
tp4_prefix512k	524,544	16.894s	7.65	3.7	7.7%

Prefix-cache UI rule Do not collapse a prefix-reuse case to a single aggregate mean. The graph must separate first/cold request from repeated/hit requests, report actual reused/computed token counts, and label the hit mechanism. The current evidence rows only expose aggregates.

Observability	Bench	TTFT ms	TPOT ms	Out tok/s
minimal	8k_c8	955.9	10.80	551.2
minimal	128k_c4	10289.9	66.77	27.2
full	8k_c8	954.0	10.70	555.4
full	128k_c4	10855.0	62.56	27.2

Observer overhead The full vs minimal observability cases should be retained as a quality-control panel. Their differences should be reported as percentage deltas with N and run-to-run variance before applying a blanket instrumentation correction.

11. Profiler & runtime reserve - redesign

Current profiler tab is not sign-off ready The page currently displays precise percentages for GPU, TP, PP, vLLM, CPU and overlap. Those values must be removed unless produced by the final profiler/timeline analyzer. In particular, a PP share cannot appear in a TP4/PP1 or TP8/PP1 single-node profile.

$$T_{\text{workload}} = A_{\text{GPU}} + B_{\text{TP}} + C_{\text{PP}} + D_{\text{PCIe/offload}} + E_{\text{vLLM}} + F_{\text{CPU/launch}} + G_{\text{other}} - O_{\text{overlap}}$$

Term	Meaning	Required source	Status type
A_GPU	KDA/MLA/MoE/GEMM/norm kernels on critical path	Nsight CUDA timeline + NVTX ranges	Timeline-attributed
B_TP	TP collectives that serialize/overlap with compute	Nsight NCCL + rank-normalized timeline; V4 NCCL only as primitive reference	Timeline-attributed / derived
C_PP	PP send/rcv + stage idle/bubble	PP>1 profile with stage/rank timestamps	Timeline-attributed
D_PCIe/offload	H2D/D2H and CPU KV offload	vLLM offload bytes/time + CUDA memcpy timeline	Measured + timeline
E_vLLM	queue, scheduler, KV block manager, admission	Prometheus + EngineCore iteration details	Directly measured where parsed
F_CPU/launch	CUDA API/driver launch/Python/Ray overhead	Nsight CPU/CUDA API timeline	Timeline-attributed
G_other	tokenization, unexplained runtime/residual	Residual after validated buckets	Unresolved/derived
O_overlap	concurrent GPU/NCCL/memcpy/runtime work	Critical-path timeline overlap	Timeline-attributed subtraction

- Do not add aggregate kernel durations across 4/8/16 ranks and call the sum wall-clock latency.

- Do not count HBM/memory traffic separately when it is already included inside GEMM/kernel elapsed time unless using a roofline model rather than a latency sum.
- Do not count cross-NUMA P2P again when it is already embodied in measured NCCL collective time.
- Do not count PP transfer and the full pipeline bubble as two independent additive terms when they overlap.
- Do not force the components to sum to TTFT/TPOT. The unexplained remainder must remain Other/Unresolved.

Pri	Profiler item	Instruction
P0	SM efficiency 74.2%	Remove unless Nsight/CUPTI metric source is present.
P0	TP overhead 21.8% / 11.2%	Remove unless computed from rank-normalized critical-path NCCL attribution.
P0	vLLM runtime 14%	Remove unless queue/scheduler/runtime ranges directly support it.
P0	Overlap 12.6%	Remove unless calculated from timeline overlap.
P0	PP 8.2%	Set N/A on PP1; only populate PP>1 runs.
P0	VRAM doughnut	Use actual Blackwell 96GB-class capacity and measured/reserved pools, not 48GB.
P0	PCIe plot	Use PCIe Gen5 and V4 measured H2D/D2H if available; badge as MEASURED-GCP-HW.
P1	Trace completeness	Show ranks expected/captured, Nsight version, NCCL trace mode, enforce-eager, warm/cold state.
P1	Phase selector	Prefill / decode / batched-decode; matched TP4 vs TP8 profile.

12. Evidence tab - make it the source of truth

Coverage finding Archived V6 single-node matrix defines 64 benchmark rows. The dashboard exposes 59 evidence rows. Six configured rows are absent (3 FP8-KV + 3 native-offload), while one extra row (tp4_closedloop_1m/c4) appears beyond the archived case file. This is a classic revision-drift signal and must be made visible.

Missing configured case	Benchmark
tp4_fp8_kv	128k_c1
tp4_fp8_kv	128k_c4
tp4_fp8_kv	512k_c1
tp4_native_offload_pressure	128k_c1
tp4_native_offload_pressure	512k_c1
tp4_native_offload_pressure	1m_c1

Extra evidence row tp4_closedloop_1m / c4 is present in the dashboard but not in the archived V6 case definition. This likely reflects a later matrix update and is useful evidence, but the UI must identify the suite revision that created it.

Pri	Evidence function	Instruction
P0	Coverage KPIs	Configured / started / completed / failed / safety-skipped / not-run, by group.
P0	Source pointer	run_id, source_file, case_manifest hash, suite_version/git SHA on every row.
P0	Actual workload	actual_input_tokens, output_tokens, sample

		count N, request_rate, max_concurrency, max_num_seqs, max_num_batched_tokens.
P0	Config state	TP/PP, GPU set, KV dtype, prefix cache, offload, chunk budget, observability profile.
P0	Multi-node	Include multi-node rows in the same evidence database or a dedicated source selector.
P1	Filters	Generate context filters from actual values; remove synthetic 32K/64K/256K choices when no run exists.
P1	Click-through	Click any chart point to open its evidence row and raw artifact path.

13. Final page architecture and required charts

Tab	Above-the-fold charts/KPIs	Architectural question
Executive	TTFT, TPOT, achieved output tok/s, queue, KV%, coverage status	Measured-only serving envelope; workload-scoped findings; data freshness.
Scale-Up	TTFT vs context; TPOT vs context; closed-loop concurrency; throughput-vs-latency Pareto; NCCL primitive	TP4/TP8 trade-off and capacity knee.
Scale-Out	128K/512K topology selector; TTFT/TPOT/tok/s; rank placement; network fingerprint; per-GPU balance	TP-vs-PP/network trade-off without universal ranking.
Long Context	1M baseline; 4K/8K/16K chunk; c1/c2/c4; per-chunk time vs accumulated context; prefix reuse	Identify regime change and long-context capacity collapse.
Scheduler & KV	KV%, running/waiting, queue, preemptions, max_num_seqs sweep, open-loop load once available	Admission/capacity mechanism, not "max users".
Profiler	Timeline, kernel/NCCL attribution, CPU/API, overlap, residual, trace completeness	Root-cause TTFT/TPOT breakdown.
Evidence	All run rows + provenance + raw paths + coverage accounting	Auditability and zero silent interpolation.

14. Mandatory annotation format below every graph

Template Observation: literal measured result. Interpretation [High/Medium/Low]: plausible mechanism. Decision: workload-scoped implication. Next evidence: trace/test needed to confirm causality. Evidence: source class + run IDs + N.

- Never write "proves" when a graph only establishes correlation.
- Never write "best" without defining the workload and objective (TTFT, TPOT, throughput, memory, cost, or network sensitivity).
- Never call closed-loop concurrency "users". If open-loop RPS exists, show offered load and achieved load separately.
- Never mix surrogate MEASURED-48B absolute performance with MODELED-K3 projections.

15. Minimum JSON schema required by the UI

Domain	Required fields
identity	run_id, suite_version, git_sha, timestamp, hostname, zone
model	model_id, revision, dtype/quantization, vllm_version,

	torch/cuda/nccl/flashinfer versions
topology	tp, pp, dp, gpu_indices, rank_to_node_gpu, numa, network_transport
workload	input_tokens_actual, output_tokens_actual, max_concurrency, request_rate, burstiness, num_prompts, sample_count
scheduler	max_num_batched_tokens, max_num_seqs, max_num_active_seqs, running, waiting, queue_time
cache	kv_cache_dtype, kv_peak_pct, prefix_hits/query/computed_tokens, preemptions, offload_bytes/time
service	ttft_mean/p50/p95/p99, tpot/itl, e2e, request_tps, input_tps, output_tps, failures
hardware	gpu_uuid, util, mem_used, power, clock; network RTT/iperf/MTU/cap; V4 primitives
profiler	profile_mode, ranks_expected/captured, kernel/NCCL timeline buckets, CPU/API, idle, overlap, residual
evidence	evidence_class, source_file, parser_version, validation_status, raw_artifact_path

16. Sign-off acceptance checklist

- ☐ No chart contains a numeric point that cannot be resolved to an evidence row or an explicitly DERIVED formula.
- ☐ No missing/not-run condition is rendered as zero.
- ☐ All 1M panels distinguish baseline c1, TP4 closed-loop c1/c2/c4, chunking, missing offload, and absent/not-configured scale-out coverage.
- ☐ TP16/PP1 and forced TP4/PP2 topology diagrams correctly reflect cross-node communication semantics.
- ☐ Every scale-out result shows rank placement and network provenance next to the performance result.
- ☐ Profiler page contains no precise decomposition percentage until trace-derived attribution is available.
- ☐ Hardware label is RTX PRO 6000 Blackwell Server Edition, 96GB GDDR7, PCIe Gen5; no Ada/48GB/Gen4 residue remains.
- ☐ Percentile charts enforce N thresholds and show N in tooltips.
- ☐ Every chart uses Observation / Interpretation / Decision / Next evidence language.
- ☐ Evidence tab displays suite revision and coverage counts by group.
- ☐ Clicking a chart point can navigate to its evidence row/raw source.
- ☐ Absolute 48B surrogate latency/tok/s is never presented as Kimi K3 performance.

17. Source register

ID	Source	Use
S1	MASTER_CHARACTERIZATION_DASHBOARD.html	Uploaded current UI, 2,523 lines; all seven tabs audited.
S2	v6_qualification_results_with_logs.zip	Raw qualification manifests, benchmark JSON, server logs, metrics and GPU telemetry; qualification values previously validated.
S3	v5_vllm_profiling_suite_v6.tar.gz	Archived V6 single-node and multi-node case definitions used for coverage comparison.
S4	NVIDIA RTX PRO 6000 Blackwell Server Edition official page	96GB GDDR7, PCIe Gen5 x16, 1597 GB/s memory bandwidth.
S5	vLLM 0.29/stable documentation	bench serve request-rate/max-concurrency; cache/offload semantics; max_num_seqs scheduler semantics.

- NVIDIA: <https://www.nvidia.com/en-us/data-center/rtx-pro-6000-blackwell-server-edition/>
- vLLM bench serve: <https://docs.vllm.ai/en/stable/cli/bench/serve/>
- vLLM cache config 0.29.0: <https://docs.vllm.ai/en/v0.29.0/api/vllm/config/cache/>
- vLLM serve scheduler arguments: <https://docs.vllm.ai/en/stable/cli/serve/>

Appendix A. Master UI remediation matrix

Tab	Pri	Change	Exact instruction
GLOBAL	P0	Replace legacy Ada/48GB/Gen4 labels	Use Blackwell Server Edition/96GB/Gen5 from detected inventory.
GLOBAL	P0	Remove hard-coded measured arrays	Bind all plots to source JSON/evidence rows.
GLOBAL	P0	Correct badge taxonomy	MEASURED-48B vs MEASURED-GCP-HW vs DERIVED/UNRESOLVED.
EXEC	P0	Replace 8/8 top-level status	V6 configured/completed/failed/gated/not-run.
EXEC	P0	Measured-only serving envelope	No synthetic contexts/concurrency cells.
SCALE-UP	P0	Measured-only context points	8K/128K/512K/1M; interpolation dashed/derived.
SCALE-UP	P0	Crossover wording	Bracket 128K-512K; fitted point derived.
SCALE-UP	P0	Matched TP4/TP8 concurrency	No synthetic TP8 c2/c4/c16/c32.
SCALE-UP	P0	NCCL evidence source	Use V4/Nsight, separate time/algbw/busbw.
SCALE-OUT	P0	Correct TP16 topology	Cross-node TP16 ring.
SCALE-OUT	P0	Correct forced TP4/PP2	TP4 per node + PP boundary, no split TP ring.
SCALE-OUT	P0	Remove universal verdict	Scope to matched context/concurrency/objective.
SCALE-OUT	P0	Remove unsupported network causality	Require raw traffic/cap/retry/queue evidence.
SCALE-OUT	P1	Add rank placement validation	Rank->node->GPU UUID->PCI->NUMA.
SCALE-OUT	P1	Add 128K/512K selector	Show NOT RUN where absent.
LONG	P0	Reword 1M feasibility	Specify exact passed matrix, not 100% blanket.
LONG	P0	Correct chunk finding	16K lowest 1M c1 TTFT among tested chunks.
LONG	P0	Correct prefix evidence	Use measured 128K/512K cold-vs-hit; 1M is derived unless run.
LONG	P0	Correct KV units	Show %, not GB, unless bytes verified.
LONG	P1	Add regime-change panel	Chunk iteration time vs accumulated context + runtime signals.
SCHED	P0	Correct KV chart values	Use evidence values; no linear extrapolation.
SCHED	P0	Gate percentile reporting	P95 N>=20; P99 N>=100.
SCHED	P1	Promote max_num_seqs sweep	512K c4 4/8/16 dedicated learning.
PROF	P0	Remove fabricated percentages	Keep equation + unresolved ledger until trace data.
PROF	P0	No PP share in PP1 profile	PP=N/A for single-node PP1.
PROF	P0	Correct VRAM/PCIe	96GB-class + PCIe Gen5.
PROF	P1	Add trace completeness	Ranks captured, trace mode, warm/cold, eager/graph.
EVIDENCE	P0	Coverage by group	Configured/started/completed/failed/gated/not-run.
EVIDENCE	P0	Show revision drift	Suite/git SHA + manifest hash.
EVIDENCE	P0	Include multi-node/profiler sources	Unified evidence layer.
EVIDENCE	P1	Click-through traceability	Chart point -> evidence row -> raw artifact.

Evidence Appendix - Source Rows & Traceability

Why this appendix exists The UI must be auditable. The following landscape pages reproduce all 59 current dashboard Evidence rows so that chart claims can be checked against the values actually embedded in the dashboard. Presence here does not replace raw-log validation.

Appendix B. Current dashboard Evidence rows (59)

Interpretation rule These are the rows embedded in the current dashboard. Their presence in the UI is not equivalent to raw-log validation. Qualification rows were previously raw-validated; full-production rows should retain source paths/run IDs so the same audit can be repeated.

Case	Bench	TP/PP	Input	Conc	TTFT ms	TPOT ms	Out tok/s	KV peak	Status
tp4_qualification	8k_c1	TP4 / PP1	8,192	c1	224.3	4.45	188.4	0.1%	COMPLETED
tp4_qualification	8k_c8	TP4 / PP1	8,192	c8	956.5	10.70	555.0	1.0%	COMPLETED
tp4_qualification	128k_c1	TP4 / PP1	131,072	c1	4534.1	5.08	24.7	1.6%	COMPLETED
tp4_qualification	512k_c1	TP4 / PP1	524,288	c1	31955.5	7.58	2.0	6.5%	COMPLETED
tp8_qualification	8k_c1	TP8 / PP1	8,192	c1	267.7	6.35	135.7	0.1%	COMPLETED
tp8_qualification	8k_c8	TP8 / PP1	8,192	c8	1036.0	13.92	445.8	0.9%	COMPLETED
tp8_qualification	128k_c1	TP8 / PP1	131,072	c1	4819.6	7.03	22.4	1.6%	COMPLETED
tp8_qualification	512k_c1	TP8 / PP1	524,288	c1	28216.8	9.47	2.2	6.4%	COMPLETED
tp4_context_baseline	8k_c1	TP4 / PP1	8,192	c1	222.3	4.45	162.4	0.1%	COMPLETED
tp4_context_baseline	128k_c1	TP4 / PP1	131,072	c1	4540.8	5.10	24.7	1.6%	COMPLETED
tp4_context_baseline	512k_c1	TP4 / PP1	524,288	c1	31978.5	7.56	2.0	6.5%	COMPLETED
tp4_context_baseline	1m_c1	TP4 / PP1	1,000,000	c1	93384.5	10.24	0.3	12.3%	COMPLETED
tp8_context_baseline	8k_c1	TP8 / PP1	8,192	c1	267.6	6.33	119.5	0.1%	COMPLETED
tp8_context_baseline	128k_c1	TP8 / PP1	131,072	c1	4824.3	7.05	22.4	1.6%	COMPLETED
tp8_context_baseline	512k_c1	TP8 / PP1	524,288	c1	28166.5	9.44	2.2	6.4%	COMPLETED
tp8_context_baseline	1m_c1	TP8 / PP1	1,000,000	c1	74850.3	12.07	0.4	12.2%	COMPLETED
tp4_prefill_focus	8k_prefill	TP4 / PP1	8,192	c1	223.0	4.38	16.9	0.0%	COMPLETED
tp4_prefill_focus	128k_prefill	TP4 / PP1	131,072	c1	4542.1	5.09	0.9	1.6%	COMPLETED
tp4_prefill_focus	512k_prefill	TP4 / PP1	524,288	c1	31960.1	7.84	0.1	6.4%	COMPLETED
tp4_prefill_focus	1m_prefill	TP4 / PP1	1,000,000	c1	93354.2	10.64	0.0	12.3%	COMPLETED
tp4_decode_focus	8k_decode_c1	TP4 / PP1	8,192	c1	225.6	4.47	204.0	0.1%	COMPLETED
tp4_decode_focus	8k_decode_c8	TP4 / PP1	8,192	c8	957.2	9.23	721.0	1.0%	COMPLETED
tp4_decode_focus	8k_decode_c16	TP4 / PP1	8,192	c16	1242.6	15.64	885.1	2.0%	COMPLETED
tp4_chunk4k	128k_c1	TP4 / PP1	131,072	c1	5235.5	5.13	11.5	1.6%	COMPLETED
tp4_chunk4k	128k_c4	TP4 / PP1	131,072	c4	11475.9	142.05	12.2	4.8%	COMPLETED
tp4_chunk4k	512k_c1	TP4 / PP1	524,288	c1	40307.2	7.57	0.8	6.4%	COMPLETED
tp4_chunk4k	1m_c1	TP4 / PP1	1,000,000	c1	122103.8	10.28	0.1	12.2%	COMPLETED
tp4_chunk8k	128k_c1	TP4 / PP1	131,072	c1	4544.5	5.11	13.2	1.6%	COMPLETED
tp4_chunk8k	128k_c4	TP4 / PP1	131,072	c4	10928.1	117.39	13.9	6.5%	COMPLETED
tp4_chunk8k	512k_c1	TP4 / PP1	524,288	c1	31966.6	7.58	1.0	6.5%	COMPLETED
tp4_chunk8k	1m_c1	TP4 / PP1	1,000,000	c1	93375.9	10.37	0.2	12.3%	COMPLETED

tp4_chunk16k	128k_c1	TP4 / PP1	131,072	c1	4366.5	5.09	13.7	1.7%	COMPLETED
tp4_chunk16k	128k_c4	TP4 / PP1	131,072	c4	10922.1	106.51	14.5	6.6%	COMPLETED
tp4_chunk16k	512k_c1	TP4 / PP1	524,288	c1	30498.9	7.54	1.0	6.6%	COMPLETED
tp4_chunk16k	1m_c1	TP4 / PP1	1,000,000	c1	89164.1	10.26	0.2	12.5%	COMPLETED
tp4_closedloop_8k	c1	TP4 / PP1	8,192	c1	224.9	4.47	187.6	0.1%	COMPLETED
tp4_closedloop_8k	c4	TP4 / PP1	8,192	c4	610.8	7.27	415.3	0.5%	COMPLETED
tp4_closedloop_8k	c8	TP4 / PP1	8,192	c8	916.5	10.73	559.7	1.0%	COMPLETED
tp4_closedloop_8k	c16	TP4 / PP1	8,192	c16	1155.8	19.27	673.2	2.0%	COMPLETED
tp4_closedloop_8k	c32	TP4 / PP1	8,192	c32	1623.7	34.96	774.4	4.0%	COMPLETED
tp4_closedloop_128k	c1	TP4 / PP1	131,072	c1	4538.8	5.11	24.7	1.6%	COMPLETED
tp4_closedloop_128k	c4	TP4 / PP1	131,072	c4	10596.0	64.37	27.2	6.5%	COMPLETED
tp4_closedloop_128k	c8	TP4 / PP1	131,072	c8	12722.5	186.98	28.0	13.1%	COMPLETED
tp4_closedloop_128k	c16	TP4 / PP1	131,072	c16	37256.6	242.85	28.2	14.6%	COMPLETED
tp4_closedloop_512k	c1	TP4 / PP1	524,288	c1	32028.5	7.56	2.0	6.5%	COMPLETED
tp4_closedloop_512k	c2	TP4 / PP1	524,288	c2	40299.2	367.40	2.0	12.8%	COMPLETED
tp4_closedloop_512k	c4	TP4 / PP1	524,288	c4	87703.8	433.95	2.0	12.9%	COMPLETED
tp4_closedloop_1m	c1	TP4 / PP1	1,000,000	c1	93460.4	10.20	0.3	12.3%	COMPLETED
tp4_closedloop_1m	c2	TP4 / PP1	1,000,000	c2	150654.3	239.25	0.3	15.5%	COMPLETED
tp4_closedloop_1m	c4	TP4 / PP1	1,000,000	c4	231503.5	267.69	0.3	15.5%	COMPLETED
tp4_512k_maxseq4	512k_c4	TP4 / PP1	524,288	c4	88007.1	433.81	2.0	12.9%	COMPLETED
tp4_512k_maxseq8	512k_c4	TP4 / PP1	524,288	c4	87994.3	433.76	2.0	12.9%	COMPLETED
tp4_512k_maxseq16	512k_c4	TP4 / PP1	524,288	c4	87974.9	433.77	2.0	12.9%	COMPLETED
tp4_prefix128k	prefix128k	TP4 / PP1	131,328	c1	910.7	5.31	80.7	1.9%	COMPLETED
tp4_prefix512k	prefix512k	TP4 / PP1	524,544	c1	16894.4	7.65	3.7	7.7%	COMPLETED
tp4_observer_minimal	8k_c8	TP4 / PP1	8,192	c8	955.9	10.80	551.2	1.0%	COMPLETED
tp4_observer_minimal	128k_c4	TP4 / PP1	131,072	c4	10289.9	66.77	27.2	6.5%	COMPLETED
tp4_observer_full	8k_c8	TP4 / PP1	8,192	c8	954.0	10.70	555.4	1.0%	COMPLETED
tp4_observer_full	128k_c4	TP4 / PP1	131,072	c4	10855.0	62.56	27.2	6.5%	COMPLETED

Appendix C. 1M and Scale-Out developer checklist

1M must remain explicit Show: TP4 c1, TP8 c1, TP4 c1/c2/c4 closed-loop, 4K/8K/16K chunking, prefill-focused. Show native-offload 1M as missing/not surfaced. Do not imply 1M multi-node coverage unless separately added and run.

Scale-Out must remain auditable For each topology show 128K/512K result availability, rank placement, GCP zone/interface/RTT/MTU/iperf, NCCL transport, TTFT/TPOT/output tok/s, per-node balance, and trace completeness. Do not call the GCP result a local 10GbE prediction.

Area	Must show	Must not claim
1M baseline	TP4 c1 + TP8 c1 exact TTFT/TPOT/output TPS/KV%	100% of all 1M serving configurations validated
1M concurrency	TP4 c1/c2/c4 TTFT + TPOT + throughput + KV%	c4 is healthy merely because it completed
1M chunking	4K/8K/16K measured TTFT; workload/SLO context	8K is universally optimal
1M prefix/offload	Measured prefix/offload rows or NOT RUN	projected cache speedup as measured
Scale-out topology	rank placement + 128K/512K metrics + network provenance	universal winner independent of workload
Network	RTT/iperf/MTU/cap/NCCL transport	VPC saturation/TCP queues without direct evidence
Profiler	timeline-attributed GPU/NCCL/CPU/overlap/residual	aggregate kernel sums as wall-clock decomposition